

4 月 2 日作业答案

问题 1 (3.2) (1) 盒子里装有 3 只黑球、2 只红球、2 只白球, 在其中任取 4 只球. 以 X 表示取到黑球的只数, 以 Y 表示取到红球的只数. 求 X 和 Y 的联合分布律.
 (2) 在 (1) 中求 $P\{X > Y\}$, $P\{Y = 2X\}$, $P\{X + Y = 3\}$, $P\{X < 3 - Y\}$.

解 (1) 按古典概型计算. 自 7 只球中取 4 只, 共有 $\binom{7}{4} = 35$ 种取法. 在 4 只球中, 黑球有 i 只, 红球有 j 只 (剩下 $4 - i - j$ 只白球) 的取法数为

$$N\{X = i, Y = j\} = \binom{3}{i} \binom{2}{j} \binom{2}{4-i-j},$$

$$i = 0, 1, 2, 3; \quad j = 0, 1, 2; \quad i + j \leq 4.$$

于是

$$\begin{aligned} P\{X = 0, Y = 2\} &= \binom{3}{0} \binom{2}{2} \binom{2}{2} / 35 = \frac{1}{35}. \\ P\{X = 1, Y = 1\} &= \binom{3}{1} \binom{2}{1} \binom{2}{2} / 35 = \frac{6}{35}. \\ P\{X = 1, Y = 2\} &= \binom{3}{1} \binom{2}{2} \binom{2}{1} / 35 = \frac{6}{35}. \\ P\{X = 2, Y = 0\} &= \binom{3}{2} \binom{2}{0} \binom{2}{2} / 35 = \frac{3}{35}. \\ P\{X = 2, Y = 1\} &= \binom{3}{2} \binom{2}{1} \binom{2}{1} / 35 = \frac{12}{35}. \\ P\{X = 2, Y = 2\} &= \binom{3}{2} \binom{2}{2} \binom{2}{0} / 35 = \frac{3}{35}. \\ P\{X = 3, Y = 0\} &= \binom{3}{3} \binom{2}{0} \binom{2}{1} / 35 = \frac{2}{35}. \\ P\{X = 3, Y = 1\} &= \binom{3}{3} \binom{2}{1} \binom{2}{0} / 35 = \frac{2}{35}. \end{aligned}$$

$$P\{X = 0, Y = 0\} = P\{X = 0, Y = 1\} = P\{X = 1, Y = 0\} = P\{X = 3, Y = 2\} = 0.$$

分布律为

Y \ X	X			
	0	1	2	3
0	0	0	$\frac{3}{35}$	$\frac{2}{35}$
1	0	$\frac{6}{35}$	$\frac{12}{35}$	$\frac{2}{35}$
2	$\frac{1}{35}$	$\frac{6}{35}$	$\frac{3}{35}$	0

(2)

$$\begin{aligned}
 P\{X > Y\} &= P\{X = 2, Y = 0\} + P\{X = 2, Y = 1\} \\
 &\quad + P\{X = 3, Y = 0\} + P\{X = 3, Y = 1\} \\
 &= \frac{3}{35} + \frac{12}{35} + \frac{2}{35} + \frac{2}{35} = \frac{19}{35}.
 \end{aligned}$$

$$P\{Y = 2X\} = P\{X = 1, Y = 2\} = \frac{6}{35}.$$

$$\begin{aligned}
 P\{X + Y = 3\} &= P\{X = 1, Y = 2\} + P\{X = 2, Y = 1\} + P\{X = 3, Y = 0\} \\
 &= \frac{6}{35} + \frac{12}{35} + \frac{2}{35} = \frac{20}{35} = \frac{4}{7}.
 \end{aligned}$$

$$\begin{aligned}
 P\{X < 3 - Y\} &= P\{X + Y < 3\} \\
 &= P\{X = 0, Y = 2\} + P\{X = 1, Y = 1\} + P\{X = 2, Y = 0\} \\
 &= \frac{1}{35} + \frac{6}{35} + \frac{3}{35} = \frac{10}{35} = \frac{2}{7}.
 \end{aligned}$$

问题 2 (3.3) 设随机变量 (X, Y) 的概率密度为

$$f(x, y) = \begin{cases} k(6 - x - y), & 0 < x < 2, 2 < y < 4, \\ 0, & \text{其他.} \end{cases}$$

(1) 确定常数 k .

(2) 求 $P\{X < 1, Y < 3\}$.

(3) 求 $P\{X < 1.5\}$.

(4) 求 $P\{X + Y \leq 4\}$.

解 (1) 由 $\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) dx dy = 1$, 得

$$\begin{aligned}
 1 &= \int_2^4 dy \int_0^2 k(6 - x - y) dx = k \int_2^4 \left[(6 - y)x - \frac{1}{2}x^2 \right] \Big|_{x=0}^{x=2} dy \\
 &= k \int_2^4 (12 - 2y - 2) dy = k(10y - y^2) \Big|_2^4 = 8k,
 \end{aligned}$$

所以 $k = 1/8$.

(2)

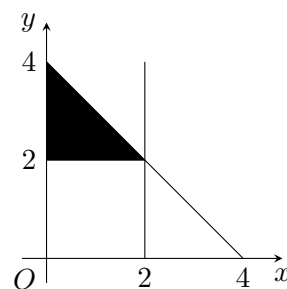
$$\begin{aligned}
 P\{X < 1, Y < 3\} &= \int_2^3 dy \int_0^1 \frac{1}{8}(6 - x - y) dx \\
 &= \frac{1}{8} \int_2^3 \left[(6 - y)x - \frac{1}{2}x^2 \right] \Big|_{x=0}^{x=1} dy \\
 &= \frac{1}{8} \int_2^3 \left(\frac{11}{2} - y \right) dy = \frac{3}{8}.
 \end{aligned}$$

(3)

$$\begin{aligned}
 P\{X < 1.5\} &= \int_2^4 dy \int_0^{1.5} \frac{1}{8}(6-x-y)dx \\
 &= \frac{1}{8} \int_2^4 \left[(6-y)x - \frac{1}{2}x^2 \right] \Big|_{x=0}^{x=1.5} dy \\
 &= \frac{1}{8} \int_2^4 \left(\frac{63}{8} - \frac{3}{2}y \right) dy = \frac{27}{32}.
 \end{aligned}$$

(4) 在 $f(x, y) \neq 0$ 的区域 $R: 0 \leq x \leq 2, 2 \leq y \leq 4$ 上作直线 $x + y = 4$ (如题 3.3 图), 并记 $G: \{(x, y) \mid 0 \leq x \leq 2, 2 \leq y \leq 4 - x\}$, 则

$$\begin{aligned}
 P\{X + Y \leq 4\} &= P\{(X, Y) \in G\} \\
 &= \iint_G f(x, y) dx dy \\
 &= \int_2^4 dy \int_0^{4-y} \frac{1}{8}(6-x-y)dx \\
 &= \frac{1}{8} \int_2^4 \left[(6-y)x - \frac{1}{2}x^2 \right] \Big|_{x=0}^{x=4-y} dy \\
 &= \frac{1}{8} \int_2^4 \left[(6-y)(4-y) - \frac{1}{2}(4-y)^2 \right] dy \\
 &= \frac{1}{8} \int_2^4 \left[2(4-y) + \frac{1}{2}(4-y)^2 \right] dy \\
 &= \frac{1}{8} \left[-(4-y)^2 - \frac{1}{6}(4-y)^3 \right] \Big|_2^4 = \frac{2}{3}.
 \end{aligned}$$



题 3.3 图

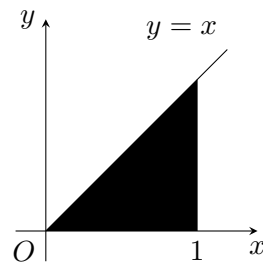
问题 3 (3.7) 设二维随机变量 (X, Y) 的概率密度为

$$f(x, y) = \begin{cases} 4.8y(2-x), & 0 \leq x \leq 1, 0 \leq y \leq x, \\ 0, & \text{其他.} \end{cases}$$

求边缘概率密度.

解 (X, Y) 的概率密度 $f(x, y)$ 在区域 $G: \{(x, y) \mid 0 \leq x \leq 1, 0 \leq y \leq x\}$ 外取零值. 如题 3.7 图有

$$\begin{aligned}
 f_X(x) &= \int_{-\infty}^{\infty} f(x, y) dy \\
 &= \begin{cases} \int_0^x 4.8y(2-x) dy, & 0 \leq x \leq 1, \\ 0, & \text{其他} \end{cases} \\
 &= \begin{cases} 2.4(2-x)x^2, & 0 \leq x \leq 1, \\ 0, & \text{其他.} \end{cases}
 \end{aligned}$$



题 3.7 图

$$\begin{aligned} f_Y(y) &= \int_{-\infty}^{\infty} f(x, y) dx \\ &= \begin{cases} \int_y^1 4.8y(2-x) dx, & 0 \leq y \leq 1, \\ 0, & \text{其他} \end{cases} \\ &= \begin{cases} \mathbf{2.4y(3 - 4y + y^2)}, & \mathbf{0 \leq y \leq 1}, \\ 0, & \text{其他.} \end{cases} \\ &= \begin{cases} \mathbf{2.4y(y - 1)(y - 3)}, & \mathbf{0 \leq y \leq 1}, \\ 0, & \text{其他.} \end{cases} \end{aligned}$$

注：在求边缘概率密度时，需画出 (X, Y) 的概率密度 $f(x, y) \neq 0$ 的区域，这对于正确写出所要求的积分的上下限是很有帮助的.